

Question

The element **X**'s simple substance reacts with oxygen at high temperature to produce a yellow solid **A**. Dissolving **A** in HClO_4 yields a perchlorate. Adding NaOH solution until neutral results in a hexanuclear cation **B** with a +6 charge in the solution. This ion possesses high symmetry and has three fourfold axes. Continuing to add NaOH produces a white solid. Adding NaOH further until alkaline and then introducing Cl_2 yields a yellow solid **C**.

The element **Y**'s simple substance can react with hot water to produce a white solid **D**, which immediately transforms into **E** containing a mass fraction of **Y** of approximately 52.35% upon contact with air.

The following statements are made:

1. The atomic numbers of **X** and **Y** are both odd numbers.
2. In the cation **B**, there exists OH^- coordinated to only one **X** atom.
3. **D** can react with solid **C** after dissolving in dilute sulfuric acid, and the sum of the coefficients on the reactant side and the sum of the coefficients on the product side of the ionic equation (with the simplest coefficients and one neutral component in both reactants and products) are both odd numbers.
4. **E** is a white solid.

What is the sum of the numbers of the correct statements?

A. 1

B. 3

C. 4

D. 5

E. 6

F. 8

G. 10

H. None of the above options are correct.

Answer

Correct Answer: **C**

Detailed Explanation

First, deduce element **X**: Combining the information (1) the simple substance reacts with oxygen at high temperature to produce a yellow solid; (2) a highly symmetric 6-core cation; (3) a yellow solid is obtained under alkaline oxidation conditions, it can be comprehensively determined that **X** is bismuth.

CHECKPOINT

1 PTS

X is bismuth

Cation **B** contains 6 Bi and carries six positive charges, indicating that it contains $6 \times 3 - 6 = 12$ OH^- . From the three four-fold axes, it can be known that it is a regular octahedral shape, and 12 OH^- are located on the 12 edges of the Bi_6 octahedron, and there is no OH^- coordinated to only one Bi atom. Statement 2 is incorrect.

CHECKPOINT

1 PTS

There is no OH^- coordinated to only one **X** atom in cation **B**

C is produced under alkaline and oxidizing conditions, and is yellow, indicating that it is NaBiO_3 .

CHECKPOINT

0.5 PTS

C is NaBiO_3

Then, deduce element **Y**: It is easy to know that **D** is a hydroxide. **E** is the product formed after **D** comes into contact with air. The phrase "immediately generated" rules out basic carbonates, and **E** is likely to be a hydroxide or a hydroxy oxide. Assume the chemical formula of **E** is $\text{YO}_a(\text{OH})_b$, try different non-negative integer combinations of a, b , it can be known that $a = 1, b = 2, \text{Y} = \text{Mn}$ is the only reasonable solution. Manganese can react with hot water, which also fits the description. From this, it is known that the atomic numbers of **X, Y** are both odd numbers, and statement 1 is correct. **E** is $\text{MnO}(\text{OH})_2$, which is a brown or black solid, and statement 4 is incorrect.

CHECKPOINT

0.5 PTS

E is a hydroxide or hydroxy oxide

CHECKPOINT

1 PTS

Y is manganese

CHECKPOINT

1 PTS

E is $\text{MnO}(\text{OH})_2$

CHECKPOINT

0.5 PTS

E is a brown or black solid

The reaction in statement 3 is the reaction of NaBiO_3 and Mn^{2+} under acidic conditions. The reaction equation is



The sum of the coefficients on the reactant side and the sum of the coefficients on the product side are both odd numbers. Statement 3 is correct.

CHECKPOINT

1 PTS



In summary, the correct statements are 1 and 3, and the answer is option C.